



Kubernetes

Stateful Set

Stateful and Stateless Apps

A stateless application does not retain any information or "state" between different requests. Every interaction with a stateless application is independent, and the app does not store any data from previous interactions.

e.g. Web Servers, Microservices APIs

A stateful application retains information, or "state," across sessions. This means that data from previous interactions is stored and can influence future requests. Stateful applications are often more complex because they rely on persistent storage to maintain their state.

e.g. Kafka, Databases like MongoDB, SQL

StatefulSet

Kubernetes is best known for managing stateless services. A StatefulSet kubernetes is a powerful concept designed to manage stateful applications, such as databases, messaging queues, and other applications that require stable network identities, unique persistent storage, and ordered deployment and scaling.

Unlike traditional stateless applications, stateful applications maintain some form of internal state or data that needs to be preserved across pod restarts or rescheduling.

StatefulSet vs Deployments

While Deployments and StatefulSets are both used for managing pods in Kubernetes, StatefulSets provide features essential for stateful applications like these:

1. Ordered Pod Management with indexed pods
2. Stable Network Identity with Headless services
3. Persistent Storage with volumeClaimTemplates

Headless Services

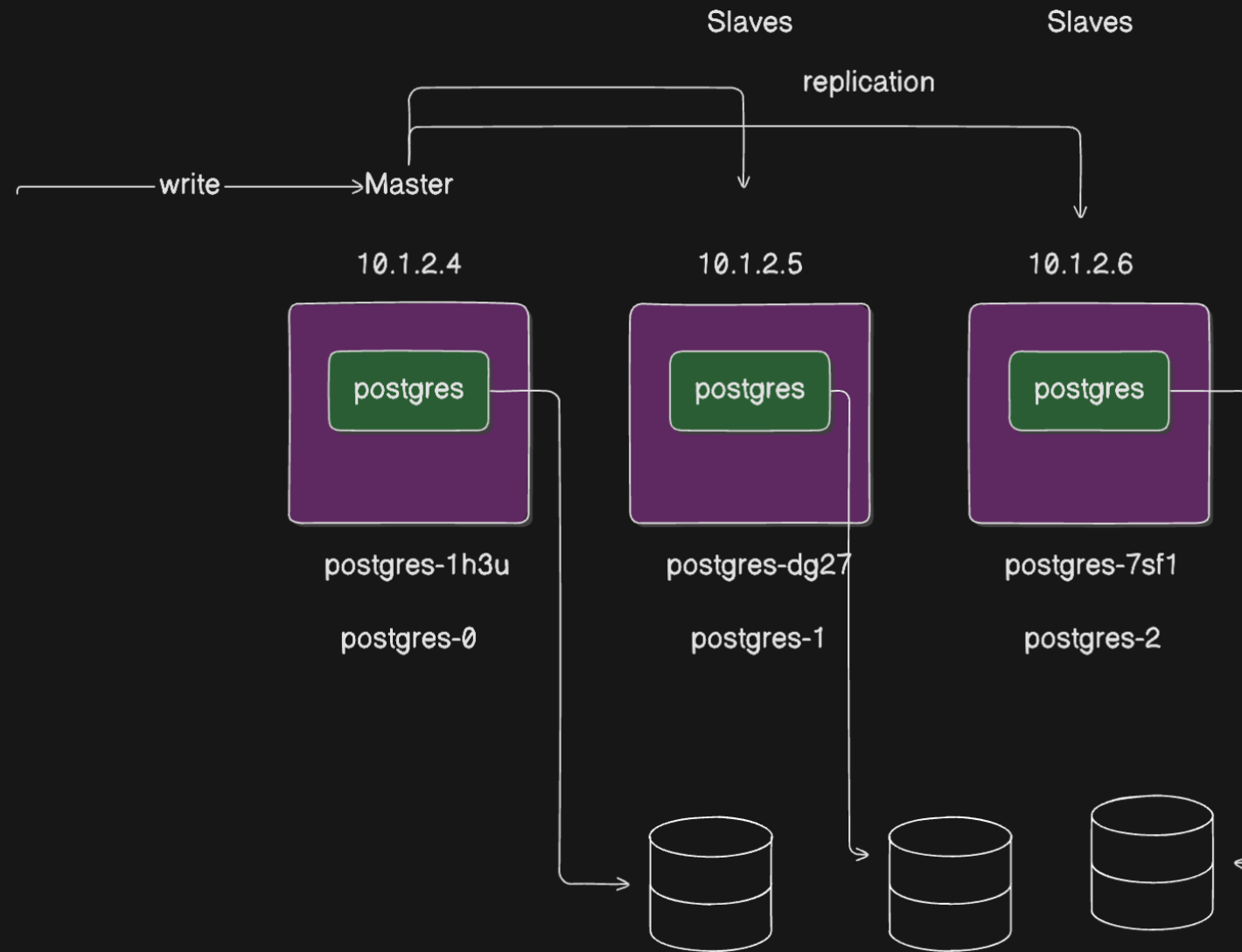
- A headless service in Kubernetes is a type of service that allows clients to connect directly to the pods associated with it, instead of going through a proxy. This is different from a regular service, which routes client requests to a pod randomly. There **is no load balancing** between Pods as with other services.
- Inter-pod communication can be done using Pod's hostnames and IP addresses.
- Endpoints are used to point the service to a set of Pods. They can be DNS names or IP addresses.
- Headless services are mainly used for clusters of applications like MongoDB, Kafka, Cassandra, etc. It can also be used to perform health check of all the pods.
- Headless services can be used for service discovery.

```
spec:  
  selector:  
    app: headless-nginx # label for the service  
  clusterIP: None # this will point directly to pods,
```

VolumeClaim Templates

- `volumeClaimTemplates` is a field within a `StatefulSet` specification that defines a template for creating `PersistentVolumeClaims` (PVCs). Each pod in the `StatefulSet` gets its own PVC based on this template.
- The primary purpose of `volumeClaimTemplates` is to ensure that each pod in a `StatefulSet` can have its own dedicated storage. It enables applications to maintain data even if the pods are deleted or rescheduled, which is critical for databases, caches, and other stateful applications.

```
volumeClaimTemplates:
  - metadata:
      name: my-storage
    spec:
      accessModes:
        - ReadWriteOnce
      resources:
        requests:
          storage: 1Gi
      storageClassName: standard
```



Stateful Set

- uniform names
- different storage solution
- headless service

